

FIRST MONEY BANK SQL QUERIES & RESULTS

Q1. Loan Summary by Type, using Loans Table

```
Query History
6
7 -- Converted loan_amount from TEXT to NUMERIC data type,
8 --to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.
9 ALTER TABLE loans
10 ALTER COLUMN loan_amount TYPE NUMERIC
11 USING loan_amount::NUMERIC;
12
13 --Q1 Loan Summary by Type
14 --Find out how many loans we gave out and the total money spent for each type of loan (Personal, Business, SME)
15 SELECT loan_type,
16 COUNT (loan_id) AS total_no_of_loans,
17 ROUND(SUM(loan_amount),2) AS total_amount_disbursed,
18 ROUND(AVG(loan_amount),2) AS average_loan_amount
19 FROM loans
20 GROUP BY loan_type
21 ORDER BY total_amount_disbursed DESC;
22
```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1

	loan_type text	total_no_of_loans bigint	total_amount_disbursed numeric	average_loan_amount numeric
1	SME	120	456500000.00	3804166.67
2	sme	69	250250000.00	3626811.59
3	personal	62	229250000.00	3697580.65
4	BUSINESS	55	211250000.00	3840909.09
5	Personal	61	187000000.00	3065573.77
6	business	52	178000000.00	3423076.92
7	Business	51	150500000.00	2950980.39
8	PERSONAL	45	147500000.00	3277777.78

Total rows: 8 Query complete 00:00:00.297

Query Result: It returns the number of loans we gave out, and the total money spent on each type of loan (Personal, Business, SME)

Q2 High-Value Defaulted Loans

Query

Query History

23

--Q2 High-Value Defaulted Loans

24

--Find the really big loans >2000000 where the customer has stopped paying back

25

SELECT loan_id, loan_type,

26

loan_amount, disbursement_date

27

FROM loans

28

WHERE loan_status = 'Defaulted'

29

AND loan_amount > 2000000

30

ORDER BY loan_amount DESC

31

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 32

	loan_id text	loan_type text	loan_amount numeric	disbursement_date text
1	LN00371	Personal	10000000	21-Sep-2024
2	LN00434	personal	10000000	2024-08-30
3	LN00477	BUSINESS	10000000	2024-11-10
4	LN00373	BUSINESS	10000000	2024-08-03
5	LN00185	SME	10000000	2024-11-13
6	LN00314	sme	10000000	2024-12-01
7	LN00486	BUSINESS	10000000	11-Jan-2024
8	LN00424	business	10000000	2024-12-13
9	LN00184	SME	10000000	2024-12-20
10	LN00341	SME	7500000	2024-03-20
11	LN00050	SME	7500000	2024-08-18
12	LN00271	BUSINESS	7500000	2024-02-11
13	LN00409	personal	7500000	2024-04-11
14	LN00324	PERSONAL	7500000	2024-04-02

Total rows: 32

Query complete 00:00:00.177

Query Result: It returns the Loan_id, loan_type, loan_amount, disbursement_date of customers with loans >2000000 where the customer has stopped paying back (Defaulted).

Q3. Lagos Region Loans with JOIN

Query

Query History

32

-- Q3 Lagos Region Loans with JOIN

33

-- Get the list of all the loans given out by branches only located in Lagos

34

SELECT l.loan_id,

35

l.loan_type,

36

l.loan_amount,

37

l.loan_status,

38

b.branch_name

39

FROM loans l

40

JOIN branches b

41

ON l.branch_id = b.branch_id

42

WHERE b.city = 'Lagos'

43

ORDER BY loan_amount DESC;

Data Output

Messages

Notifications

<

Query Result: It returns all the individual loans given specifically by branches located in the city of Lagos, showing the loan ID, type, amount, loan status, and the exact name of the branch that handled it.

Q4. Total Loans per Branch with Default Rate

Query

Query History

```
45 -- Q4 Total Loans per Branch with Default Rate
46 -- Get the branches that has the highest percentage of bad loans compared to the total loans they gave out.
47 -- Using CTE
48 WITH Branch_loans AS(
49 SELECT b.branch_name,
50 b.city,
51 COUNT (l.loan_id) AS total_loans,
52 SUM (l.loan_amount) AS total_loan_amount,
53 COUNT (CASE WHEN l.loan_status = 'Defaulted' THEN 1 END) AS number_of_defaulted_loans
54 FROM branches b
55 JOIN loans l
56 ON b.branch_id =l.branch_id
57 GROUP BY b.branch_name,
58 b.city)
59
60 SELECT branch_name, city, total_loans, total_loan_amount,
61 number_of_defaulted_loans,
62 ROUND((number_of_defaulted_loans * 100.0) / total_loans, 1) AS default_rate
63 FROM Branch_loans
64 WHERE total_loans > 5
65 ORDER BY default_rate DESC;
```

66

Data Output

Messages

Notifications

Showing rows: 1 to 10

Page 1

	branch_name text	city text	total_loans bigint	total_loan_amount numeric	number_of_defaulted_loans bigint	default_rate numeric	
1	Sabon Gari	Kano	43	164500000	10	23.3	
2	Wuse II	Abuja	26	81750000	5	19.2	
3	Bompai	Kano	47	165500000	8	17.0	

Total rows: 10

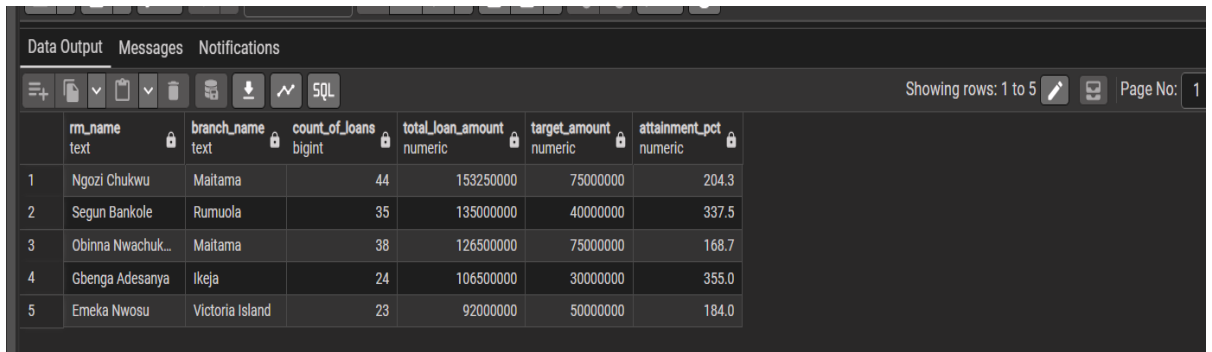
Query complete 00:00:00.548

Query Result: It returns a summary for every single branch (with more than 5 loans), showing the total number of loans they gave out, the total money spent, how many of those loans failed, and the exact percentage of their loans that ended up in default.

Q5 Top 5 Relationship Managers

```
--Q5 Top 5 Relationship Managers
--Find the 5 best staffs based on the total money they loaned out (Active and Fully paid only)
WITH RM_Performance AS (
SELECT s.rm_name, b.branch_name,
COUNT (l.loan_id) AS count_of_loans,
SUM (l.loan_amount) AS total_loan_amount,
s.target_amount
FROM staff s
JOIN branches b
ON s.branch_id = b.branch_id
JOIN loans l
ON s.rm_id = l.rm_id
WHERE l.loan_status = 'Active' or l.loan_status = 'Fully Paid'
GROUP BY s.rm_name, b.branch_name,s.target_amount
)

-- To Calculate the attainment percentage
SELECT rm_name, branch_name,
count_of_loans, total_loan_amount,
target_amount,
ROUND((total_loan_amount / target_amount) * 100, 1) AS attainment_pct
FROM RM_Performance
ORDER BY total_loan_amount DESC
LIMIT 5;
```



The screenshot shows a database query result interface. At the top, there are tabs for 'Data Output', 'Messages', and 'Notifications'. Below the tabs is a toolbar with various icons for file operations and a 'SQL' button. The main area displays a table with 7 columns: 'rm_name' (text), 'branch_name' (text), 'count_of_loans' (bigint), 'total_loan_amount' (numeric), 'target_amount' (numeric), and 'attainment_pct' (numeric). The table contains 5 rows of data, representing the top 5 relationship managers based on total loan amount. The interface also shows 'Showing rows: 1 to 5' and 'Page No: 1'.

	rm_name text	branch_name text	count_of_loans bigint	total_loan_amount numeric	target_amount numeric	attainment_pct numeric
1	Ngozi Chukwu	Maitama	44	153250000	75000000	204.3
2	Segun Bankole	Rumuola	35	135000000	40000000	337.5
3	Obinna Nwachuk...	Maitama	38	126500000	75000000	168.7
4	Gbenga Adesanya	Ikeja	24	106500000	30000000	355.0
5	Emeka Nwosu	Victoria Island	23	92000000	50000000	184.0

Query Result: It returns the top 5 relationship managers who gave out the most money in successful loans (Active & Fully paid), showing the total amount they disbursed, their specific branch, and what percentage of their set goal they achieved.

Q6: Monthly Repayment Trend

Query

Query History

105

--Converted payment_date from TEXT to DATE data type,

106

--to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.

107

ALTER TABLE repayments

108

ALTER COLUMN payment_date TYPE DATE

109

USING payment_date::DATE;

110

111

--Q6: Monthly Repayment Trend

112

--How much money does the bank collect from loan repayments each month, and people's payment status time.

113

SELECT

114

TO_CHAR(payment_date, 'YYYY-MM') AS year_month,

115

SUM(amount_paid) AS total_amount_collected,

116

COUNT(CASE WHEN payment_status = 'On-Time' THEN 1 END) AS on_time_payments,

117

COUNT(CASE WHEN payment_status = 'Late' THEN 1 END) AS late_payments,

118

COUNT(CASE WHEN payment_status = 'Missed' THEN 1 END) AS missed_payments,

119

ROUND(((COUNT(CASE WHEN payment_status = 'On-Time' THEN 1 END) * 100) / COUNT(repayment_id), 1) AS on_time_rate

120

FROM repayments

121

GROUP BY TO_CHAR(payment_date, 'YYYY-MM')

122

ORDER BY year_month ASC;

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 47

Page No: 1

	year_month text	total_amount_collected numeric	on_time_payments bigint	late_payments bigint	missed_payments bigint	on_time_rate numeric
1	2024-01	5344387	23	8	8	58.0
2	2024-02	13466542	55	18	11	65.0
3	2024-03	22413032	88	23	12	71.0
4	2024-04	26448739	110	20	23	71.0
5	2024-05	30995364	111	38	25	63.0
6	2024-06	36215266	130	43	35	62.0

Total rows: 47 Query complete 00:00:00.181

Query Result- It returns a month-by-month trend over the year, showing the total amount of money collected each month, and the exact count of payments that were made on time, late, or missed.

Q7: Customer Segment Default Analysis (3-table JOIN)

```

123
124 --Q7: Customer Segment Default Analysis (3-table JOIN)
125 -- Determine which intersection of customer demographics and geographic regions carries the highest credit risk.
126 -- Using CTE
127 WITH Customer_Risk AS(
128 SELECT c.customer_segment,b.region,
129 COUNT(l.loan_id) AS total_loans,
130 SUM(l.loan_amount) AS total_loan_amount,
131 COUNT(CASE WHEN l.loan_status = 'Defaulted' THEN 1 END) AS defaulted_loan_count
132 -- 3-Tables JOIN to link loan status to customer profiles and branch locations
133 FROM loans l
134 JOIN customers c
135 ON l.customer_id = c.customer_id
136 JOIN branches b ON l.branch_id = b.branch_id
137 GROUP BY c.customer_segment,b.region)
138
139 SELECT customer_segment,region,
140 total_loans,total_loan_amount,
141 defaulted_loan_count,
142 ROUND((defaulted_loan_count * 100) / total_loans, 1) AS default_rate
143 FROM Customer_Risk
144 WHERE total_loans >= 3
145 ORDER BY default_rate DESC;
146
147 --The results show us exactly which customer segment and region is causing the bank to lose the most money due to unpaid loans.
148
149

```

Data Output Messages Notifications							Showing rows: 1 to 24		Page No: 1
	customer_segment text	region text	total_loans bigint	total_loan_amount numeric	defaulted_loan_count bigint	default_rate numeric			
1	CORPORATE	South-Sou...	7	20000000	2	28.0			
2	Retail	North	28	84500000	8	28.0			
3	Corporate	North	31	96750000	7	22.0			
4	corporate	North	33	136500000	7	21.0			
5	retail	South-Sou...	21	70000000	4	19.0			
6	CORPORATE	North	33	145750000	6	18.0			
7	SME	South-West	16	46250000	3	18.0			
8	retail	South-West	6	26250000	1	16.0			
9	sme	South-Sou...	6	21000000	1	16.0			
10	RETAIL	North	39	122000000	6	15.0			
11	CORPORATE	South-West	21	88750000	3	14.0			
12	RETAIL	South-Sou...	7	26000000	1	14.0			
13	sme	South-West	22	89250000	3	13.0			
14	retail	North	32	86000000	4	12.0			
15	Corporate	South-West	17	74250000	2	11.0			
16	SME	South-Sou...	27	70250000	3	11.0			
17	corporate	South-West	19	69000000	2	10.0			
18	Retail	South-Sou...	14	61000000	1	7.0			
19	sme	North	28	79500000	2	7.0			
20	SME	North	30	124250000	2	6.0			
21	RETAIL	South-West	19	75250000	1	5.0			
22	Retail	South-West	7	24250000	0	0.0			
23	Corporate	South-Sou...	3	3250000	0	0.0			
24	corporate	South-Sou...	8	23500000	0	0.0			
Total rows: 24 Query complete 00:00:00.405									

Query Result: it returned a summary of type of customer (Corporate, Retail, SME) and their region, showing the total number of loans, the total money disbursed, the number of defaulted loans, and the calculated default percentage for each specific group.

Key Insight: The results show us exactly which customer segment and region is causing the bank to lose the most money due to unpaid loans.

Q8: Altering Tables

```
-- Converted Loan_amount from TEXT to NUMERIC data type,  
--to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.  
  
ALTER TABLE loans  
ALTER COLUMN loan_amount TYPE NUMERIC  
USING loan_amount::NUMERIC;  
-- Converted target_amount from TEXT to NUMERIC data type,  
--to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.  
  
ALTER TABLE Staff  
ALTER COLUMN target_amount TYPE NUMERIC  
USING target_amount::NUMERIC;  
  
--Converted amount_paid from TEXT to NUMERIC data type,  
--to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.  
  
ALTER TABLE repayments  
ALTER COLUMN amount_paid TYPE NUMERIC  
USING amount_paid::NUMERIC;  
  
--Converted payment_date from TEXT to DATE data type,  
--to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.  
ALTER TABLE repayments  
ALTER COLUMN payment_date TYPE DATE  
USING payment_date::DATE;  
|
```

Query result: Altered loans table, staff table and repayments table to correct from TEXT to NUMERIC data type and from TEXT to DATE data type, to resolve errors and allow aggregation functions like SUM and ROUND to execute correctly.